

REVIEW ARTICLE

System for Observing Play and Leisure Activity in Youth: A Systematic Review of US and Canadian Studies

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ABSTRACT

BACKGROUND: The System for Observing Play and Leisure Activity in Youth (SOPLAY) has assisted in providing valid and reliable data of youth physical activity and characteristics specific to environmental contexts. The review aimed to examine empirical research that employed the SOPLAY instrument to measure physical activity in leisure-based activity environments in North American countries.

METHODS: The review was conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses. A systematic search was conducted with an exhaustive process using 10 electronic databases to locate peer-reviewed studies implementing SOPLAY that were published between 2000 and 2021.

RESULTS: A total of 60 studies were included in the review. Most studies ($n = 35$) reported physical activity findings related to contextual characteristics measured using SOPLAY. Interestingly, a few of the studies ($n = 8$) found the provision of equipment and supervision, particularly adult supervision, significantly increased child physical activity observed.

CONCLUSIONS: This review provides information relative to the group-level physical activity observed across multiple contexts (ie, playgrounds, parks, recreation centers) using a validated direct observation instrument.

Keywords: physical activity; measurement; youth; play spaces; environmental characteristics.

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BACKGROUND

Physical activity (PA) is essential to an individual's growth, development, and health. As such, PA is promoted widely across North America, particularly within the United States.¹ PA for children is often encouraged and occurs within different environments, such as structured (eg, physical education, sports) and unstructured opportunities (eg, recess, free play). Meanwhile, most of what is directly known about PA in school-based contexts results from the direct observations of physical education classes.^{2,3} Beyond physical education, much less is known about youth PA in noncurricular settings. Therefore, PA researchers developed innovative methods to objectively capture

and quantify PA and movement patterns of children and adolescents in leisure-type contexts. Along with device-measured data (eg, accelerometry, heart rate monitors), direct observation instruments using momentary time sampling, event-, duration- and interval recording techniques have been shown as effective methods for obtaining objective PA data.^{4,5} Consequently, the System for Observing Play and Leisure Activity in Youth (SOPLAY) instrument was developed and validated⁴ with an established protocol, observer training and assessment materials (available online; <https://activelivingresearch.org/>)⁶ to evaluate PA across multiple leisure settings.⁷

SOPLAY is a widely used tool to observe and measure group-level PA, however, little is known

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about how the established protocols have been used. Such information is critical to comparing youth PA patterns in both structured and unstructured school and community-based settings, however, reliable and feasible methods of collecting behavioral data while engaged in group-level PA opportunities (eg, recess playgrounds) is limited. A review of studies implementing SOPLAY may provide key stakeholders with valuable information needed to make evidence-based decisions related to PA policy and programming and it allows researchers to create more effective methods aimed to improve school health. In addition, it is imperative to review studies implementing direct observation tools, such as SOPLAY to understand more about child and adolescent PA behavior within different context and those of different gender and race/ethnic backgrounds.

This review builds on earlier reviews that used the System for Observing Fitness Instruction Time (SOFIT)^{3,8} in physical education classes and the System for Observing Play and Recreation in Communities (SOPARC)⁹ in park and community settings. Along with other direct observation tools designed for use outside of physical education context, instruments such as the System for Observing Children's Activity and Relationship during Play (SOCARP)¹⁰ and the Bedimo-Run Assessment Tool—Direct Observation (BRAT-DO)¹¹ provide researchers with different options to evaluate PA. The current study contributes to the body of knowledge associated with direct observation of play and leisure-type PA and complements both the SOFIT and SOPARC reviews. Surveillance data have been shown valuable for increasing and improving advocacy efforts and direct observations are needed to establish databases relative to the development of databases relative to PA participation in leisure-type settings in the United States. It also provides information on the utility and generalizability of SOPLAY and identifies where additional research on program and protocol accountability may be needed. The primary purpose of this review study was to evaluate the overall findings associated with using SOPLAY to observe and measure group-level PA in school- and community-based settings across North America from the development of the tool in 2000 until 2021. A secondary aim was to review the methods and consistency of reports regarding the implementation of SOPLAY.

METHOD

The review followed guidelines outlined in the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) checklist^{12,13} and the Cochrane Handbook for Systematic Reviews of Interventions.¹⁴ The following sections summarize the research methods, discuss main findings, and provide

recommendations for future studies using SOPLAY and implications for school health.

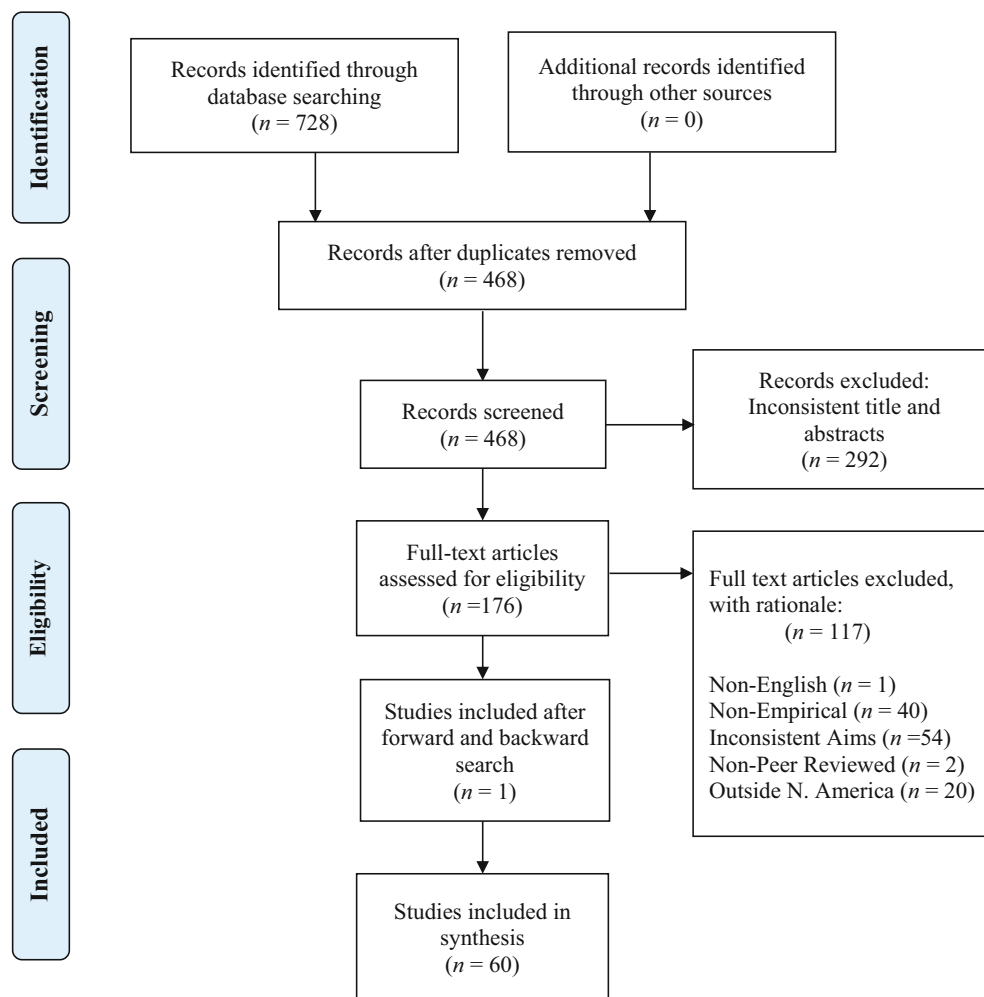
Eligibility Criteria

Systematic searches were conducted primarily by 2 research team members, while other members initiated the review and guided the development of assessment criteria and search strategy. The PICOS framework guided the review process. Further, only empirical, English language, peer-reviewed articles examining children and adolescent PA using SOPLAY in North America qualified for inclusion. Moreover, the (P) population of interest included only youths from North America; (I) intervention was considered any study employing SOPLAY in school- and community-based locations; (C) comparator were any studies lacking administration of SOPLAY; (O) outcomes included reported PA, contextual characteristics, and methodological decisions; and lastly, (S) study type focused on only empirical, peer-reviewed studies. The research team intentionally selected studies examining preadolescents and adolescents residing in North American countries due to translational challenges and stark differences in educational and PA policy in countries outside of North America. Government reports, position statements, methodology articles, commentaries, abstracts, conference proceedings, book chapters, unpublished dissertations and theses, nonrefereed articles, and reports beyond North America were excluded.

Search Strategy and Study Selection

To select pertinent papers, a specified keyword search was conducted in the following 10 databases of 2 research-intensive university electronic libraries: (1) Academic Search Ultimate Plus, (2) Education Full-Text (Wilson), (3) ERIC Educational Literature (Ebsco), (4) Sports Medicine and Education Index, (5) APA PsycInfo, (6) PubMed, (7) ScienceDirect, (8) Scopus, (9) SPORTDiscus (Ebsco) and (10) Web of Science. Ten databases were used to expand the search scope by including as many records as possible during the identification phase. Two reviewers independently conducted searches between September and November in 2021 and they completed an updated search in March 2022 to ensure all articles published to that date were included. Searches employed an algorithm that included the possible combinations of the keywords: SOPLAY, "System for Observing Play and Leisure Activity in Youth" AND parks, recreation, recess, outdoors, schools. Mesh terms were applied to the appropriate terms, for example "out*" was integrated into the algorithm for databases, such as PubMed to further expand the search to include all terms beginning with "out." Each search consisted of including the above-mentioned

Figure 1. PRISMA flowchart outlining the stepwise search strategy



algorithm using all fields to include titles, abstracts, and keywords (Data S1).

Database filters (ie, English, Date [2000-2021], Full-Text, and Peer-Reviewed) were applied to confine the results so that the appropriate eligibility criteria were established prior to the search. First, titles and abstracts of the records retrieved during the initial search were reviewed and evaluated on a 3-point criterion: 0 = not consistent, 1 = partially consistent, and 2 = consistent. Articles received 4 points if title and abstract was consistent with the selected PICOS for this review, 3 points if title was consistent and abstract was partially inconsistent and vice versa, and 2 or below if title and abstract of record was partially inconsistent or consistent with stated PICOS. All irrelevant papers were excluded after screening the titles and identifying the abstracts. After the initial selection, full-text articles were screened using the same criterion against all 7 inclusion/exclusion criteria. All records that scored at or above 11 out of 14 were included. Throughout

the process, the 2 assessors maintained high interrater reliability (>90%) and any disagreements regarding article eligibility were deliberated until arriving to consensus. Figure 1 provides an overview of the identification, screening, and inclusion process.

Data Extraction and Analysis

All references identified from the search were compiled and saved into both a Google Excel spreadsheet and Zotero reference management software to ensure storage and proper backup files for subsequent data extraction procedures. Following title and abstract screening, each study was examined to determine the extent to which inclusion criteria were met. Two investigators independently extracted data from studies, which included: (1) authors, (2) publication year, (3) article type, (4) examination of contextual characteristics, (5) sample demographics, (6) setting or context, (7) use of SOPLAY to examine PA (sedentary, walking, MVPA, and very active), and (8) methodology

used (ie, research design, observation time-periods, average days observed, number of scanned areas, type of observer training, and reported observer reliability). All data were compiled into a Microsoft Excel Google Sheet for ease of collaboration and to track article eligibility status. To ensure consistency throughout the coding process, half the articles ($n = 30$) were double coded by 2 investigators. Investigators responsible for reviewing and extracting data met regularly to check for consistency throughout the data extraction process to discuss and resolve any disagreements. Inter-coder reliability assessments for the 30 articles was high ($>90\%$).

Supplementary data were recorded relative to the primary components of the instrument, including: number of observed individuals, contextual characteristics (ie, accessible, useable, supervised, organized, equipment provided), day of the week (ie, weekdays or weekends), time of day (ie, before school, lunchtime, or afterschool), demographics (eg, age, sex, race/ethnicity), and activity levels by sex and location (eg, playgrounds, parks, schools). In addition, the research methods used (ie, quantitative and mixed methods), geographic region in North America (ie, Canada, Mexico, and United States), and other frequently reported variables (eg, deviations from the original SOPLAY protocol, observer training, and reliability reporting) were extracted if available. Descriptive statistics were applied to demographics, such as number of individuals, gender, race/ethnicity, age, observation period, and frequency of observation.

RESULTS

From the 10 databases used, 728 records were initially identified (Figure 1). Following removal of duplicate records ($n = 260$), we screened 468 records and excluded those with inconsistent titles and those with only abstracts available ($n = 292$). After screening 176 records for eligibility, 117 were excluded. This resulted in 59 records meeting the predetermined inclusion criteria. An additional study was located during forward- and backward-searching of refereed references, therefore a total of 60 studies were reviewed.

Research Designs and Physical Activity

The earliest study that used SOPLAY reported PA in free-play contexts that included playgrounds, recess, and before and after-school programs.⁷ In our review, 54 studies (93.3%) used a quantitative approach, 5 employed a mixed-method, and 1 used a qualitative research design.¹⁵ The research designs were diverse, with most being quasi-experimental designs or direct observational methods only (96.5%).

Setting and context for observations varied, with 32 studies occurring on school playgrounds of which 30

(94%) assessed recess behavior (Table 3). A few studies took place in out-of-school locations such as After-School Programs ($n = 4$) and Summer Day Camps ($n = 4$). The remaining studies examined leisure time PA across various time periods in school, with 3 studies focusing on elementary¹⁵⁻¹⁷ and 4 on secondary school populations.^{7,18-20} Only 8 studies (13.3%) were conducted in local parks.

Most (98.3%) studies examined PA at the group-level PA using the SOPLAY protocol, but 1 assessed PA using the related SOFIT procedure.²¹ Nearly all (83.3%) studies used the standard SOPLAY protocol,²² however, 9 (15%) had ambiguous modifications. A majority (81.4%) of papers that examined leisure PA reported sedentary (S), walking (W) and very active (V) observations and reported mean proportions/percentage, mean PA counts, odds ratios, and various regression modeling techniques. Over two-third (68.3%) reported children's group-level PA engagement. In addition, 69.5% of studies reported group-level PA based on gender differences; however, only 3 studies²³⁻²⁵ were stratified by racial or ethnic grouping.

Roughly 60% (35 of 60) examined rates of MVPA, but considerable differences existed in how SOPLAY activity codes were combined or interpreted. For instance, 17 of the 35 studies reviewed combined the codes "W" and "V" to be considered as MVPA; however, the study by Gustat et al.²⁶ did not report MVPA results. In contrast, the activity code "V" was considered MVPA for 8 of the 35 studies that reported MVPA. Some reports of studies ($n = 12$) were unclear about how SOPLAY activity codes were combined to represent MVPA.

With the available SOPLAY PA data, the following conditions were considered. Across the review, only 8 studies identified rates or proportions of children observed in S, W, or V behavior within school playground settings, more specifically during recess. Of these, several were removed from our analysis due to concerns about reporting modifications to SOPLAY. On average, children engaged in $62.1 \pm 16.6\%$ MVPA during recess periods when the aggregated standard SOPLAY activity codes "W" and "V" for estimates of MVPA were used (Figure 2). Males engaged in MVPA an average of $65.3 \pm 13.3\%$ of the time with females averaging $58.7 \pm 13.5\%$. Figure 3 presents the 5 studies that used the activity code "V" as the marker for MVPA to examine children's PA during recess (Figure 3). In these 5 studies, average MVPA was $41.2 \pm 12.3\%$; males ($32.1 \pm 16.8\%$) engaged in more MVPA than females ($25.5 \pm 18.2\%$).

Few studies (only 5 of 59) examined adolescents during leisure-time PA opportunities at different time-periods in middle school (beyond recess; Figure 4). Combining codes "W" and "V," males engaged in less MVPA ($16.5 \pm 22.5\%$) than females ($19.3 \pm 18.2\%$).

Figure 2. Results of reported children's mean MVPA percentage accumulated during recess from studies that combined SOPLAY activity codes "W + V" to represent MVPA

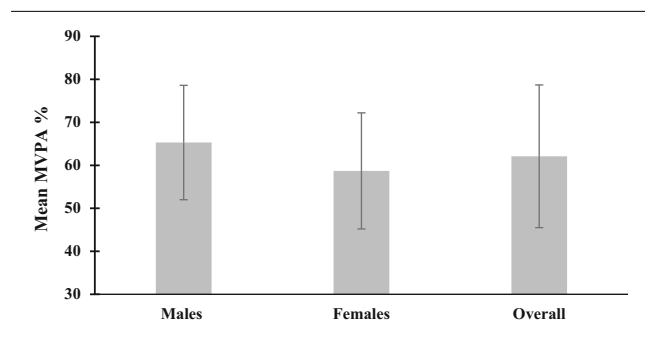
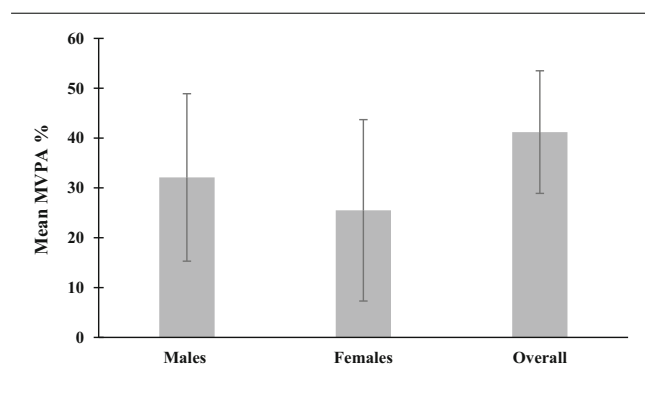


Figure 3. Results of reported children's mean MVPA percentage accumulated during recess from studies that clearly identified the use of the SOPLAY activity code "V" to represent MVPA

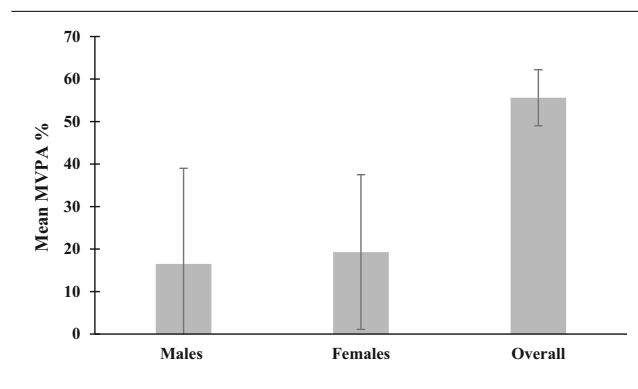


The overall average MVPA for adolescents in middle-school leisure physical activity opportunities was 55.6% across the 5 studies. Overall, most studies (96.6%) found that PA, particularly MVPA, varied by gender and contextual characteristics. The provision of equipment and supervision, particularly adult supervision during PA opportunities was shown to be significantly related to increased MVPA as reported across 8 studies.²⁷⁻³² In contrast, 2 studies found no significant change in PA with the presence of supervision or inclusion of equipment.^{33,34}

Participants and Settings

Of the 60 studies analyzed, 75% (n = 45) reported the number of individuals observed, and 43.3% indicated the proportion of participants by gender and 58.3% identified them by race and/or ethnicity (Table 1). The number observed in studies ranged from 61 to 36,995 ($\bar{X}$ = 5240.9, SD = 8767.2), with 88.9% of studies including fewer than 10,000 participants (Table 2). Fifty-seven articles identified participants' age grouping. Of these 57 studies, 68.3% included

Figure 4. Studies reporting adolescent's mean MVPA percentage during outside of school contexts from studies that combined SOPLAY activity codes "W + V" to represent MVPA



school-aged children and only 7% included both children and adults.

Of the 60 studies identified, 57 (95%) were conducted in the United States and 3 (5%) in Canada^{29,43,44}; no studies took place in Mexico. Of the US studies, 21 (36.8%) were from the West, 9 (15.8%) from the Midwest, 9 (15.8%) from the South, 4 (7%) from the Northeast, and 4 took place in both the midwestern and southern regions (7%) (Table 2).

Additionally, most studies (68.3%) were conducted in school playground recess settings, 8 in parks, and 28 in areas categorized as "other." Additionally, 35 studies assessed the associations of contextual characteristics with PA participation.

In terms of contextual characteristics, 35 studies assessed associations of them with PA participation. Few studies reported the percentages of the contextual characteristics observed; 15.6% reported accessibility, 9.4% usability (ie, good condition, lighting), and approximately 40% reported whether areas were supervised or equipped. Thirty-five percent of the 60 studies (n = 21) examined the relationships of participant PA with contextual characteristics or defined target areas. Of those, approximately half (47.6%) modified the contextual characteristics of the original instrument to include additional features (eg, area size, playground equipment type, social interactions, sidewalk conditions).

Features of Observations Using SOPLAY

The number of observation days reported (n = 40 studies) ranged from 1 to 200, with 37 studies (92.5%) observing fewer than 72 days. The majority of observations were conducted outdoors (56.7%) and on weekdays (68.3%) (Table 1 and 2). Most observations occurred during school hours or afterschool (78.3%), with an exception during which observations were made strictly before school.⁴⁵ Twenty-five studies reported observation periods as "other" which

Table 1. Characteristics of Reviewed Studies (N = 60)

Variables	Study n	%	$\bar{X}$	SD
# of individuals observed	45	75.0	5240.89	8767.24
Female	26	43.3	3715.00	5229.98
Male	26	43.3	4213.73	5885.57
Race/ethnicity	35	58.3	-	-
African American/Black	27	45.0	-	-
Asian American	11	18.3	-	-
Caucasian/White	26	43.3	-	-
Hispanic/Latino	26	43.3	-	-
Native American/Pacific Islander	7	11.7	-	-
Mixed	5	8.3	-	-
Other	11	18.3	-	-
Age				
Elementary (K-5th/6th)	35	58.3	-	-
Middle (6th/7th-8th)	10	16.7	-	-
Elementary and middle (K-8th)	6	10.0	-	-
High (9th-12th)	1	1.7	-	-
Elementary, middle and high (K-12th)	1	1.7	-	-
Adults	0	0	-	-
Children and adults	4	6.7	-	-
Not mentioned	3	5.0	-	-
Observation period				
Before-school	1	1.7	-	-
During-school	25	41.7	-	-
After-school	8	13.3	-	-
2 or more	14	23.3	-	-
Other	12	20.0	-	-
Observation days of week				
Weekday	41	68.3	-	-
Weekend	5	8.3	-	-
Both	12	20.0	-	-
Not mentioned	2	3.3	-	-
Days observed	40	66.6	33.8	44.39
Area setting				
Indoor	41	68.3	-	-
Outdoor	5	8.3	-	-
Both	12	20.0	-	-
Not mentioned	2	3.3	-	-
Number of scans	10	16.7	6212.30	13,723.86

included weekends and summer breaks. Altogether, only 13.3% of the studies specified the type of recording device used to identify the number observed in S, W, and V. Most (87.5%) used a manual mechanical counter and 12.5% used electronic devices (eg, iPad). Only 28.3% reported the temperature during observation sessions.

Observer Training and Interobserver Reliability

Only half the studies reported how observers were trained. In these, 56.7% used in-person training, 26.7% used online training (eg, recorded videos created by McKenzie⁶) and 16.7% used both in-person and online training. Of all reviewed studies, 76.7% presented evidence of interobserver reliability during data collection. In general, the percent agreement and correlation coefficient exceeded .85 and .70, respectively.

DISCUSSION AND CONCLUSION

This review follows the standard PRISMA model to examine studies that used SOPLAY to assess PA and relevant contexts in schools and other settings in North America. Using PRISMA to search for papers and extract data should enable additional researchers to replicate the review process. Findings from this review may assist others in conducting quality SOPLAY studies and data extraction processes. We identified 60 papers that met our inclusion criteria, in which complements prior systematic reviews of the SOFIT and SOPARC instruments^{3,8,9} and contributes to the understanding of PA within play and leisure contexts, especially during unstructured settings such as school recess and parks.

Findings Across SOPLAY Studies

SOPLAY is widely used in North America, with most studies taking place in the United States. Advantages of the SOPLAY instrument include its ability to reliably obtain PA data at the group level while simultaneously assessing contextual data both within and beyond schools and among individuals of all ages. Our findings are generalizable since SOPLAY has been used across diverse settings.

Play and leisure physical activity. Most studies (98.3%) reported measures of PA using SOPLAY. Nonetheless, assessing PA using additional or alternative instruments was also used, with 16.7% of studies including individual-level assessments (eg, accelerometry, pedometers, heart-rate monitors) and 15.3% of studies using self-report surveys (ie, Physical Activity Resources Assessment [PARA]⁴⁸), individual interviews, and other direct observation tools (ie, SOFIT, SOPARC, SOCARP, SOSPAN). Again, only 35 of the 60 studies reported MVPA using the SOPLAY instrument protocol which assesses group-level PA.

The studies using standard SOPLAY protocol varied in how they reported PA (S, W, V) and MVPA. Only a few papers evaluated differences in PA by gender, activity type, and contextual characteristics. In terms of observing recess PA using SOPLAY, results across the studies included in this review reported higher MVPA percentages in comparison studies evaluating recess PA using different methods. Using accelerometers, Ariz et al⁷⁶ found children engaged in roughly 20% of MVPA during school recess which significantly differed from the approximate 60% depicted in Figure 2. Perhaps the method of combining activity codes “W” and “V” to represent MVPA with SOPLAY observations provides inflated percentages of time spent in moderate or vigorous intensities when compared to other objectively derived measures. On the other hand, SOPLAY findings pertaining to sex differences in PA were similar to recently published articles which observed boys participating in higher

Table 2. Characteristics Reported Among SOPLAY Studies (2000-2021)

Author(s), (Year)	State/ Province	Total Days Observed	Monday- Friday	Sunday- Saturday	# of Observations	# of Scans	Total # Observed	(K, ICC, IOA)
Anthamatten et al. ³⁵	NY	2		x	288	-	545	IOA > 80
Anthamatten et al. ³⁶	CO	36	x	x	261	-	-	IOA = 87
Anthamatten et al. ²⁷	CO	8	x	x	106	-	-	IOA = 87
Baker et al. ³⁷	MO	4	x		31,069	-	9900	IOA > 80
Barnas and Ball ³⁸	MO	1	x		8870	-	8870	-
Barnas et al. ³⁹	MO	10	x		-	-	433	K = .90
Beemer et al. ⁴⁰	MI	10	x		-	-	364	K = .85
Beets et al. ⁴¹	-	-	x		294	-	292	K = .76-.88
Beets et al. ⁴²	SC	-	x		2005	-	580	-
Behrens et al. ²⁸	SC	27			4649	-	2462	-
Belley-Ranger et al. ⁴³	CO	6	x		1148	-	-	-
Berg ⁴⁴	QC	-	x		267	-	-	-
Black et al. ⁴⁵	BC	-	x		2268	-	215	-
Bleeker et al. ⁴⁶	NV	10	x		-	-	618	IOA > 80
Bocarro et al. ⁴⁷	-	145	x		341	-	1573	IOA > 80
Bocarro et al. ⁴⁸	-	153	x		1188	-	6735	IOA = 94.2
Brink et al. ⁴⁹	-	-	x		1188	-	6735	IOA = 94.2
Chin and Ludwig ⁵⁰	CO	36	x		5488	-	1185	-
Colabianchi et al. ⁵¹	NY	-	x		1339	-	510	ICC = .72
Colabianchi et al. ⁵²	OH	-	x	x	100	-	-	ICC = .71-.96
Coleman et al. ²¹	OH	-	x	x	200	-	-	IOA > 90
Coughenour et al. ²³	KS	6	x		297	-	144	*
Dyment et al. ²⁹	NV	8	x	x	40	-	1423	IOA > 85
Elder et al. ⁵³	ON	18	x		-	43	457	IOA > 80
Escaron et al. ³⁰	CA	-	x		-	2592	23,466	IOA > 80
Escaron et al. ⁵⁴	CA	57	x		11,574	-	5909	IOA > 80
Farley et al. ⁵⁵	CA	57	x		9340	-	-	*
Farley et al. ⁵⁶	LA	9	x	x	-	-	1465	IOA = .96
Findholt et al. ⁵⁷	LA	62	x	x	-	-	-	ICC > .95
Floyd et al. ²⁴	OR	2	x		-	-	142	IOA > 80
Floyd, Suau et al. ²⁵	FL & IL	-	x	x	-	-	9454	*
Frost et al. ⁵⁸	FL & IL	-	x	x	-	-	9454	IOA = .79-.97
Grant et al. ⁵⁹	CO	3	x		22	-	148	IOA > 90
Gustat et al. ²⁶	MT	24	x		-	-	61	r = .89
Huberty et al. ⁶⁰	LA	18	x	x	-	-	-	-
Huberty et al. ⁶¹	NE	12	x		1022	-	888	IOA = 95.9
Hughey et al. ⁶²	-	5	x		1082	-	667	IOA > 80
Khorana et al. ⁶³	-	4	x		4648	-	500	IOA = 84-100
Long et al. ⁶⁴	NY	28	x		-	-	-	-
Lorenz et al. ¹⁸	NY	-	x		317	-	5882	-
Loukaitou-Sideris and Sideris ⁶⁵	-	3	x		-	-	1452	IOA > 80
Mayfield et al. ⁶⁶	CA	200	x	x	-	-	1245	K = .82-.97
McKenzie et al. ⁷	SC	-	x		-	1766	-	ICC = 0.74
McKenzie et al. ¹⁶	CA	72		x	-	-	25,944	IOA > 80
Moore et al. ¹⁷	CA	65	x		-	2359	36,995	-
Raney et al. ⁶⁷	-	35	x		-	-	-	-
Robinson et al. ³⁴	CA	-	x		-	-	437	IOA = 98.2
Saint-Maurice et al. ⁶⁸	MS	-	x		-	-	97	-
Saint-Maurice et al. ⁶⁹	NE	3	x		-	382	100	r = .82-.92
Sallis et al. ¹⁹	-	-	x		-	624	160	-
Sallis et al. ²⁰	CA	72	x		-	-	25,944	IOA = 90
Smith et al. ¹⁵	CA	-	x		72	-	26,616	r = .99
Spengler et al. ⁷⁰	UT	30	x		-	44,627	-	ICC > 80
Stanton-Chapman et al. ⁷¹	FL & IL	-		x	-	-	3410	IOA > 80
Suau et al. ⁷²	GA	59	x	x	-	416	2296	ICC = .90-.98
Weaver et al. ⁷³	FL & IL	-		x	-	-	9514	K = .79-.97
Yu et al. ⁷⁴	SC	-	x		-	8528	800	ICC = .88-.98
Zarrett et al. ³¹	AZ	13		x	-	-	514	IOA > 80
Zarrett et al. ³²	-	35	x		-	786	-	r = .97
	SC	4			76	-	484	IOA = 98

Studies used varied statistics to assess observer reliability. Intraclass correlation coefficient (ICC) and interobserver agreement (IOA) are presented in percentages, and Cohen's Kappa are presented in "K." Columns with asterisks (*) indicate that multiple kappa values were reported. Studies that used Pearson Correlation used *r*.

Table 3. Summary of physical activity findings specific to SOPLAY measures

Author(s), Publication Year	Sample and Setting	Setting, Research Design, Condition, and/or Stratification	SOPLAY PA Measures			Contextual Characteristics and Target Areas	Other PA Measures
			Rates of PA		MVPA		
			S, W, V				
Agaronov et al. ²⁵	Children (n = 545)	Interactive Children's Museum • Observational (Direct) • 2 weekend observation days; Winter 2013	Modified SOPLAY instrument • Examined mean rates of MVPA (%) as proportion of all children observed	V = MVPA • Males: 37.6%; females: 32.8%; all: 35.2% • No significant gender differences in MVPA participation	-	-	
Anthamatten et al. ³⁵	Children 6-11 years	School playgrounds (recess) • Quasi-experimental design: Compared utilization of the school yard playgrounds (LL, non-LL, and control) • 4 weekday observation days; 3 time periods (before-, lunchtime, and after-school)	Standard SOPLAY activity codes (S, W, V) • Examined mean rates of MVPA (%) as proportion of all children observed	^a Mean MVPA = 37.9%	• When all periods (before-school, lunchtime, and after-school) are taken into consideration, there was significantly greater utilization of LL schoolyards	-	
Anthamatten et al. ²⁷	Children 5-11 years	School playgrounds (recess) • 9 elementary schools • Quasi experimental design Compared 3 old LL, 3 new LL, and 3 control schools • Observation days occurred from 2005 to 2006	Standard SOPLAY activity codes (S, W, V) • Compared and examined mean rates of MVPA (%) as proportion of all children observed and rates of MVPA (%) by playground zone type	^a Males: 39.63% ^a Female: 32.90% ^a All: 37.90% • Boys MVPA (%) significantly higher in play field areas, hard-surface play areas and play equipment areas	• Utilization was significantly higher for boys in play field and basketball areas and higher for girls in play equipment areas • Swing areas had highest rate of MVPA (66.6%). Unprogrammed grass areas without equipment had lowest rates (4.4%)	-	
Anthamatten et al. ³⁶	Children 4-12 years (N = 9900) 67.1% FRL	School playgrounds (recess) • 24 elementary schools • Compared renovated, non-renovated, and control schools, and vertical/horizontal playground features • Observed 12 schools in Spring 2009 and Spring 2010	Standard SOPLAY activity codes (S, W, V) • Compared and examined mean rates of MVPA (%) as proportion of all children observed and utilization of playground zone type (horizontal and vertical)	^a Males: 39.7 ± 0.4% ^a Females: 37.3 ± 0.4% ^a All: 38.5 ± 0.3%	• Significant positive associations were observed between utilization and feature density among all children ($\beta = .20$; $p < .001$) and a statistically significant but small association was observed between MVPA and feature density for girls ($\beta = 12$; $p < .013$)	-	

Table 3. (Continued on next page)

Author(s), Publication Year	Sample and Setting	Setting, Research Design, Condition, and/or Stratification	SOPLAY PA Measures			Contextual Characteristics and Target Areas	Other PA Measures
			Rates of PA		MVPA		
			S, W, V				
Baker et al. ³⁷	Children 1st-4th grade (N = 8870)	School playgrounds (recess) 4 elementary schools - Observational (direct) - Observations occurred across 3 seasons from Spring 2013- Spring 2014; 1 observation day each school during 3 time periods (before-, lunchtime, and after-school)	Standard SOPLAY activity codes (S, W, V) - There was a decline in sedentary students from Spring 2013-2014 across all 4 schools - Females: W and V behavior increased between Spring 2013- Spring 2014 ($p < .001$); - Males: W and V slightly declined between Spring-Fall 2013 but increased by Spring 2014 ($p < .001$)	^a MVPA (%) differed between schools	Schools installed walking track and implemented PA brain breaks; however SOPLAY PA was not reported on features	Qualitative interviews focused on PA policy	
Barnas and Ball ³⁸	Children 3-5th grade (N = 433)	School playgrounds (recess) - Experimental design: - Zoned vs. Traditional playground - Compared zone 1, zone 2 and control playground 3 obs. weeks total; 2 weeks, 10 days recess participated in PE activities	Standard SOPLAY activity codes (S, W, V) - Compared changes in children observed in sedentary and Active (%) in each activity zone (Zone 1 & Zone 2) % Sedentary Control: Males: 35.4%; females: 45.3%; all: 40.0% Zone 1: Males: 42.1% females 42.1%; all: 35.4% Zone 2: Males: 47.8%; females: 47.8%; all: 39.7%	W + V = Active - Control: Males: 64.6%, females: 54.9%; all: 59.9% - Zone 1: Males: 70.3% females 58.1%; all: 64.6% - Zone 2: Males: 67.3%, females 52.4%; all: 60.5%	- For girls, there was a nonsignificant trend in increased observed PA of 3.2% ($p = .06$) in the ZP Week 1 and dropping below baseline observations in ZP Week 2 ($p = .13$). - For boys, there was a significant increase of 5.9% of observed PA in ZP Week 1 ($p < .001$) and remained significantly elevated by 3% in ZP Week 2 ($p = .03$)	Accelerometer: GT3X-BT	
Barnas et al. ³⁹	Children 3rd-5th grade (N = 364)	School playgrounds (recess) 2 elementary school playgrounds - Compared utilization and participation of children in zoned and traditional playgrounds 10 observations days; 2 observations per recess	Standard SOPLAY activity codes (S, W, V) - S + W = Low or no activity - Traditional playgrounds, 79.4% of observed students participated in S or W and 20.6% in V - Zoned playgrounds, 30.6% of students were observed in V, with the students observed in S or W decreasing to 69.4% ($p = .0001$)	V = Vigorous	-	Pedometer: Walk4life Neo II	

Table 3. (Continued on next page)

Author(s), Publication Year	Sample and Setting	Setting, Research Design, Condition, and/or Stratification	SOPLAY PA Measures			Contextual Characteristics and Target Areas	Other PA Measures
			Rates of PA		MVPA		
			S, W, V				
Beemer et al. ⁴⁰	Children 8-13 years 3-6th grade (N = 292) 80% FRL	Classroom PA 20-week classroom-based PA gamification intervention: pre- and post-assessment	Standard SOPLAY activity codes (S, W, V) - Compared and examined children's mean % rates of sedentary and MVPA (pre- and post-gamification) - There was a significant increase in girls and boys MVPA participation - During an activity break between the last week of the standard and gamified interventions	V = MVPA - There was a significant effect of intervention, with a 27.3% increase in student engagement in MVPA during an activity break during the gamified intervention weeks compared with the standard intervention weeks ($p = 0.03$) - During the gamified intervention, MVPA increased 4.9% per week ($p > 0.05$)	-	-	
Beets et al. ⁴¹	Children (N = 580) 4.3-13.1 years	Afterschool programs (ASPs) 12 ASPs - Quasi-experimental design (baseline vs post) - Compared proportion of PA (S, W, V) across various ASP characteristics - Observations between Fall 2010 and Spring 2011	Standard SOPLAY activity codes (S, W, V) - Declines in the children observed S (%) and increases in W and VPA (%) were found across time - Baseline (S/W/V) Males: 50.1/31.4/18.2% Females: 62.1/24.0/13.8% Post assessment (S/W/V) Males: 39.0/36.5/24.5% Females: 51.1/31.0/17.9%	-	- From baseline to post assessment, decreases in the % of staff verbally promoting PA and doing other ASP activities, with increases in the number of staff observed supervising. - Decrease in the use of organized activities	-	
Beets et al. ⁴²	Children (N = 2462) K-5th grade ≤12 years 25-35% FRL	Summer day camps (SDCs) - Direct observation 27 observation days - Weekday observations (Mon-Fri) 10:1 staff-to-student ratio - Observations occurred during Summer 2011	Standard SOPLAY activity codes (S, W, V) - Examined the proportion (%) of children engaged in each activity category (S, W, V) and across modified contextual characteristics - Males (S/W/V) 74.2/16.4/9.4% Females (S/W/V) 79/13.5/7.5	-	- Across all observations, time allocated for PA opportunities was the most prevalent (38% of SDC schedule), followed by enrichment and drop-off/pick-up times - For activity context variables, the presence of equipment and scheduled free-play time was associated with a decrease in the proportion of boys and girls observed sedentary	-	

Table 3. (Continued on next page)

Author(s), Publication Year	Sample and Setting	Setting, Research Design, Condition, and/or Stratification	SOPLAY PA Measures			Other PA Measures
			Rates of PA		Contextual Characteristics and Target Areas	
			S, W, V	MVPA		
Behrens et al. ²⁸	Children 1st-8th grade	School playgrounds (recess) 2 school districts - Quasi-experimental design (Baseline vs follow-up) - Structured playgrounds compared to playgrounds with low, moderate, and large amounts of equipment 3 days in the Fall; 3 days in the Spring	Standard SOPLAY activity codes (S, W, V) - Compared proportion (%) and time (min) children were engaged in S, W, MVPA, and V - Statistically significant changes included increases in both male and females vigorous % for schools equipped (low, moderate, large) - For structured schools, no significant increase in moderate or vigorous activity	W + V = MVPA - Baseline Structured 72.9% Equipped 68.3% - Follow-up Structured 68.1% Equipped 72.3%	- Equipped and structured - Low equipment led to more sedentary for boys' sedentary % - Moderate and large equipment provisions led to more MVPA % ($p < .004$) - Low Equipment MVPA % 68.5; Moderate equipment MVPA % 71.2; large equipment MVPA % 71.4	-
Bailey-Ranger et al. ⁴³	Children, adolescent, adults—Individuals w/disabilities (N = 47)	Physical education class 15 elementary schools, 6 high schools, and 1 specialized (adult) - Case studies (n = 15) - Observations conducted Sept. 2012-April 2013	Modified SOPLAY activity codes for students with disabilities and environments - Adapted PA levels examined by levels of fun, type of disability and level of social interaction - PA not reported	-	-	Semi-structured interviews focused on PA enjoyment & peer interactions
Berg ⁴⁴	Children 3-5 years Pre-K (N = 145)	School Playground-Recess 4 preschool playgrounds - Schools selected based on differences in playground design and structure	- Modified SOPLAY activity codes - Children's Activity Rating Scale (Puhl et al., 1990) - Observation for Recoding PA in Children—Preschool Version (Brown et al., 2006) - Across all 4 playgrounds, children were engaged in sedentary for 50.6%, moderate 29%, and vigorous 20%	-	-	-

Table 3. (Continued on next page)

Author(s), Publication Year	Sample and Setting	Setting, Research Design, Condition, and/or Stratification	SOPLAY PA Measures		Contextual Characteristics and Target Areas	Other PA Measures
			S, W, V	Rates of PA		
Black et al. ⁴⁵	Children 3rd grade	School playgrounds (recess) 2 elementary playgrounds - Cross-sectional design 1 school implemented a Jogging and Walking Program (JAWS) compared to non-JAWS school - Observations occurred over 9 weeks	Standard SOPLAY activity codes (S, W, V) - JAWS (S/W/V) Males: 13/59/28% Females: 12/56/22% All: 14/62/17% - non-JAWS (S/W/V) Males: 53/30/16% Females: 49/35/16% All: 58/25/17% - Control (S/W/V) Males: 59/30/11% Females: 56/31/17% All: 62/30/8%	W + V = MVPA On non-Jaws days, 50% participated in MPA at both schools, however, on non-JAWS days, boys were more physically active, with 68% participating in MVPA at school B	- The highest populated areas for both schools on non-JAWS days were the general blacktop areas; however, approximately 50% of the children in these areas were sedentary. - At school B on days when the JAWS program was offered, the highest populated area was the JAWS track, and 99% of those children participated in MVPA	SOPARC
Bleeker et al. ⁴⁶	Children 4th-5th grade (N = 1573)	School playgrounds (recess) 29 schools from 6 U.S. cities; 17 treatment, 10 control 6 structured recess observations over 4 or 5 days - Cohort 1: Spring 2010-2011, Cohort 2: Spring 2011-2012	Standard SOPLAY activity codes (S, W, V) Mean % of students who were observed in moderate or vigorous activity, and vigorous activity	W + V = MVPA - Control Males: 70.9% Females: 64.1% - Treatment Males: 67.8% Females: 59.7% - No significant differences were found in the mean % Traditional playgrounds - Zoned playgrounds, 30.6% of students were observed participating in V, and statistically significant ($p = .0001$) of girls or boys observed engaging in MVPA or vigorous activity	Both girls and boys in the treatment group were more likely than their control group counterparts to engage in playground games (4-square, tetherball, dodge ball, wall ball, Simon says) and less likely to engage in sedentary activities	Accelerometer: GT3X
Bocarro et al. ⁴⁷	Adolescents (N = 6735)	Interscholastic Program and Varsity Sports 4 middle schools - Observation occurred from April 2009 and Sept-Dec 2009 during the weekdays (Mon. through Thu.) - Seasons lasted 8-10 weeks	Standard SOPLAY activity codes (S, W, V) - Examined PA by sport: policies (intramural versus varsity) physical settings, and supervision - Intramurals (S/W/V) 46.7/32.6/20.9% - Varsity Sport 54.2/29.8/17.8% - Overall 52.4/29.8/17.8%	^a No significant supervision by policy interaction effects on the odds of engaging in MVPA were observed in girls. The effects of supervision on the likelihood of engaging in MPA somewhat depended on the activity setting.	Most children in intramural schools were observed in gyms (68.6%), followed by multi-purpose fields (12.2%), tracks (8.4%), and baseball fields (6.4%). In varsity schools, children were observed in multi-purpose field areas (31.8%), gyms (28.8%), tracks (15.5%), and baseball fields (8.4%)	EER (kcal/kg/m²)

Table 3. (Continued on next page)

Author(s), Publication Year	Sample and Setting	Setting, Research Design, Condition, and/or Stratification	SOPLAY PA Measures				Other PA Measures
			Rates of PA		Contextual Characteristics and Target Areas		
			S, W, V	MVPA			
Bocarro et al. ⁴⁸	Adolescents (N = 6735)	Interscholastic Program & Varsity Sports 4 middle schools - Direct observation - Observation occurred from April and Sept–Dec. 2009 - Weekday observations (Mon–Thu)	Standard SOPLAY activity codes (S, W, V) - Examined PA by sport policies (intramural vs varsity) physical settings, and supervision - Metabolic Equivalents (METs) and final logit model for level of PA across sports - Regression models found a significant interaction effect between sport delivery model and gender.	-	Three sports (soccer, track, and basketball) had a significant positive association with increased PA, regardless of sport delivery model.	METs	
Brink et al. ⁴⁹	Children 4–11 years PreK–5th grade (N = 5488)	School playgrounds (recess) 9 elementary schools - Quasi-experimental design: 3 established, 3 recently built, and 3 control 6 schools were learning landscapes (LL) programs 4 observation days at each school	Standard SOPLAY activity codes (S, W, V) % of students engaged in Sedentary compared to Active - W + V = Active - Recently Built (S/active) 33.9/70.0% - Established (S/active) 35.7/70% - Control (S/active) 34.6/59.9%	-	- Boys' and girls' activity rates were significantly greater on LL schools' soft surface structured areas than control environments - In comparison with control schools, the % of active boys was dramatically higher on LL schools' hard surface unstructured areas (49.6% vs 63.3%), % of active girls was slightly but not significantly lower (56.2% vs 54.0%)	EER (kcal/kg/m²)	
Chin and Ludwig ⁵⁰	Children (N = 1339) 88.33% FRL (non-REP) 81.90% FRL (REP schools)	School playgrounds (recess) 25 elementary playgrounds: 15 recess enhancement program (REP), 10 non-REP schools - Quasi-experimental design: Control (non-REP) vs. Intervention (REP w/o Coach vs. REP w/ Coach) 2 to 4 observation days	Standard SOPLAY activity codes (S, W, V) - non-REP (S/W/V) 37.8/35.2/37.8% - REP w/o coach 37.1/23.8/38.8% - REP w/ coach 35.6/23.9/41.1%	-	- Contextual characteristics examined across the 3 types (REP1, REP2, non-REP) - Supervision and equipment	-	
Colabianchi et al. ⁵¹	Children, adults	20 School playgrounds 10 renovated vs. 10 unrenovated playgrounds 10 observation days per school area during outside of school hours	Standard SOPLAY activity codes (S, W, V) Proportion of active, moderately active, and vigorously active	^a Males: 60.0 ± 0.34% ^a Females: 49.0 ± 0.33% ^a All: 40.0 ± 0.41%	- Boys more vigorously active at renovated playgrounds - Girls more moderately active at unrenovated playgrounds when compared to girls at the renovated playgrounds	-	

Table 3. (Continued on next page)

Author(s), Publication Year	Sample and Setting	Setting, Research Design, Condition, and/or Stratification	SOPLAY PA Measures			Contextual Characteristics and Target Areas	Other PA Measures
			Rates of PA				
			S, W, V	MVPA			
Colabianchi et al. ⁵²	Children, adults	20 School playgrounds • 10 renovated vs. 10 unrenovated playgrounds • 10 observation days per school area during outside of school hours	Standard SOPLAY activity codes (S, W, V) • Examined the association between attributes and utilization of playgrounds and the proportion of individuals active (%)	a52% of the persons observed on the playgrounds were moderately or vigorously active with more boys than girls classified as active (60% vs 49%) and adults the least active (40%)	-	-	
Coleman et al. ²¹	Children 3rd-6th grade (N= 144)	1 ASP • Observation design (direct) • 6 observation days per ASP (Mon, Tue., and Thu.) • 3 Fall/3 Spring weekday observations	• SOFT activity codes were used to assess environmental support	• MPPA reported, however, authors used SOFT instead of SOPLAY	-	-	
Coughenour et al. ²³	Children, adolescents ≤18 years (N = 1423)	10 Parks • Direct observation • Social and environmental evaluation • Weekend and weekday observation (Jun-Jul. 2012)	Standard SOPLAY activity codes (S, W, V) • Males (S/W/V) 17.5/36.7 /45.8% • Females 25.6/39.9 /34.5%	-	• PA reported as odds ratios • Males were more likely to be observed walking and vigorous when compared to sedentary	PARA	
Dymont et al. ²⁹	Children K-8th grade	School playgrounds (recess) • 2 primary schools; 1 Canadian school (7 target areas) and 1 Australia school (6 target areas) • Observations occurred over 7 days in April 2007 • School ground greening initiatives	Standard SOPLAY activity codes (S, W, V) • Examined PA across 7 different target areas of recess playgrounds and how intensity differs by gender across selected target areas • The green area encouraged the highest percentage of children present to be engaged in moderate physical activity (51%)	-	• At the Canadian school, the treed grassy berm (42%) and the treed concrete steps (43%) encouraged the highest % of sedentary behavior, followed by the open asphalt (34%) • Vigorous activity was highest in manufactured equipment (42%) and the paved sporting court (33%)	-	

Table 3. (Continued on next page)

Author(s), Publication Year	Sample and Setting	Setting, Research Design, Condition, and/or Stratification	SOPLAY PA Measures				Contextual Characteristics and Target Areas	Other PA Measures
			Rates of PA		MVPA	S, W, V		
Elder et al. ⁵³	Children 5-7 years Low-income ($< \$30,000/\text{year}$)	13 School playgrounds (recess) - Quasi-experimental design: (Baseline and 1-year follow-up) 3 observations periods (before-, lunchtime, after-school) over 3-month period	Standard SOPLAY activity codes (S, W, V) - Changes in mean PA proportion by contextual characteristics (supervised/organized/equipped) - No significant treatment-by-time interactions for % S, W, V - Baseline (S/W/V) Males: 28.1/38.6/28.6% Females: 36.7/34.6/33.4% - Follow-up Males: 33.9/35.4/31.0% Females: 40.9/28.2/30.7%	W + V = MVPA - No significant treatment-by-time interactions for % MVPA - Boys, there was a significant decrease in the % of children observed in MVPA in the intervention group only - Baseline- Control male: 70.8% Control female: 65.1% Intervention male: 71.9% Intervention female: 63.3% - Follow-up Control males: 67.4% Control females: 62.9% Intervention male: 66.1% Intervention female: 63.3%		-	-	
Escaron et al. ³⁰	Children, adolescents 86-100% FRL	School playgrounds (recess) 14 elementary and 5 secondary schools - Observations occurred in 3 waves from Fall 2015-2016 across 3 randomly selected days in 2-weekday observations during time periods (before-, lunchtime, after-school)	Standard SOPLAY activity codes (S, W, V) - Low proportions of students engaged in vigorous activity (16.4%) relative to walking (46.6%) and sedentary activity (37%) during leisure periods - Boys were more engaged in VPA than girls (18.6% to 13.2%)	W + V = MVPA Overall MVPA: 63% Male: 66.8% Female: 57.5%	- Playground equipment was associated with higher rates of MVPA, particularly among elementary boys ($p < .0001$). - MVPA was significantly higher in areas with equipment among elementary school girls ($\chi^2 = 33.3$, $p < .0001$), elementary school boys ($\chi^2 = 44.1$, $p < .0001$), secondary school girls ($\chi^2 = 15.5$, $p < .0001$), and among secondary school boys ($\chi^2 = 5.16$, $p = .02$).	-	-	
Escaron et al. ⁵⁴	Children, adolescents 79-100% FRL	School playgrounds (recess) 5 school districts; 19 schools; 14 elementary and 5 secondary schools - Quasi-experimental design: (pre- and post-assessment) - Collected PA data Fall and Spring 2015-2017. 3 weekday observations per target area during 3 time periods (before-, lunchtime, after-school)	Standard SOPLAY activity codes (S, W, V) - Males (S/W/V): 33.2/48.2/18.6% - Females: 42.5/44.3/13.2% - All: 37.0/46.6/16.4%	W + V = MVPA The mean proportion of males engaged in MVPA increased significantly from 26.2% (38.6) at baseline to 33.2% (42.6) at follow-up, while females engaged in MVPA increased significantly from 20.3% (35.5) at baseline to 25.6% (39.5) at follow-up	The proportion of areas with supervision present was only SOPLAY characteristic that was found to increase from baseline to follow-up	-	-	

Table 3. (Continued on next page)

Author(s), Publication Year	Sample and Setting	Setting, Research Design, Condition, and/or Stratification	SOPLAY PA Measures				Other PA Measures
			Rates of PA		Contextual Characteristics and Target Areas		
			S, W, V	MVPA			
Farley et al. ⁵⁵	Children 2nd-8th grade	School playgrounds (recess) 2 low-income areas - Quasi-experimental design (intervention vs comparison neighborhood) - Observations occurred after-school from Apr. 2003-May 2005	Standard SOPLAY activity codes (S, W, V) - W + V = Active 33% S, 33% W and 33% V - Overall, 30% more active children in the intervention neighborhood (50.4%) compared with the comparison neighborhood (38.7%)	-	-	Self-report PA surveys	
Farley et al. ⁵⁶	Children 2nd-8th grade 37% of household headed by females	School playgrounds (recess) 2 low-income neighborhoods - Direct observation 2-year observation of 1 schoolyard 46 weekday observation days; 16 weekend observation days from May 2003 through April 2005	Standard SOPLAY activity codes (S, W, V) - S: 33% - W: 33% - V: 33% - No significant differences among play areas on percent of children observed walking or active	W + V = Active Active: 66%	- Children more likely to be very active in equipped areas. - Children playing in the installed play structure area were nearly twice as likely as those in the field to be coded as "very active" (51% vs 25%, $p < .001$)	-	
Findholt et al. ⁵⁷	Children, adolescents, adults (N = 142)	School playgrounds (recess) 4 elementary - Mixed methods design: Community-based participatory research (CBPR) - Observations (Mar. 2007-May 2008)	Standard SOPLAY activity codes (S, W, V) On average, 23% of enrolled elementary students were sedentary at recess and only 16% participated in vigorous activity during the observed periods	-	School C (29.1%), F (19.3%) and H (19.3%) reported the highest very active %	PARA Semi-structured interviews	
Floyd et al. ²⁴	Children, adolescents, adults (N = 9456)	28 Parks 10 Tampa, 18 Chicago - Direct observation - Observations (Mar.-Jun. 2005) - Weekends (Fri.-Sun)	Standard SOPLAY activity codes - Unclear how instrument was modified, however activity codes reported (S, W, V) - Overall (S/M/W) 65/24/11 - Results varied across cities: Chicago (S/M/W) 51/28/22% - Tampa 70/22/8%	-	Higher temperature, lower shade, unorganized activity, child age group, lower income, Hispanic ethnicity and males were significantly associated with moderate intensity activity in Tampa	-	

Table 3. (Continued on next page)

Author(s), Publication Year	Sample and Setting	Setting, Research Design, Condition, and/or Stratification	SOPLAY PA Measures			Other PA Measures
			Rates of PA		Contextual Characteristics and Target Areas	
			S, W, V	MVPA		
Floyd, Spengler, Maddock, Gobster and Suau ²⁵	Children, adolescents, adults (N = 9456)	28 Parks • 10 Tampa, 18 Chicago • Direct observation • Observations (Mar.-Jun. 2005) • Weekends (Fri.-Sun.)	Modified SOPLAY activity codes • Unclear how instrument was modified, however activity codes seem to align (S, W, V) • Overall (S/W/V) 65/23/11 • Results varied across cities: • Chicago (S/W/V) 51/28/22% • Tampa 70/22/8%	-	-	-
Frost et al. ⁵⁸	Children (N = 148) 5th-6th grade 77.7% FRL	School playgrounds (recess) • Quasi-experimental design: baseline, post-assessment 1, and post-assessment 2 • May 2014-May 2015 • Observed twice each recess period	Standard SOPLAY activity codes (S, W, V) • Baseline (S/W/V) 49.7/37.8/ 12.6% • Post-assessment 1 26.4/34.7/38.8% • Post-assessment 2 32.5/32.5/21.7%	V = MVPA • Baseline: 12.6% • Post-assessment 1: 38.8% • Post-assessment 2: 21.7% • Significant change in the % of students engaged in MVPA from baseline to the 1-year follow-up (<i>p</i> < .05) • % of children engaged in MVPA increased 33.1% from May 2014- May 2015	-	-
Grant et al. ⁵⁹	Children 3rd-6th grade (N = 61)	School playgrounds (recess) • American Indian Tribal commu- nity • Experimental design: pre- and post-assessment; facilitator, and non-facilitator-led activities; CBPR • Observations (Oct.-Dec. 2013) during weekdays (Mon.-Thu.)	Standard SOPLAY activity codes (S, W, V) • Examined mean PA counts dur- ing baseline and intervention; Stratified PA by demographics and 3 activity zones • When combined, males and females reveal a significant decrease in sedentary activity in Zone 2 (<i>p</i> < .01). • Separately, males and females reveal a significant decrease in sedentary activity for both genders (males, <i>p</i> < .01; females, <i>p</i> < .01) in Zone 2	W + V = MVPA • Significant increases in MVPA (<i>p</i> < 0.05) were seen in Zone 1 • Overall, Zone 2 reported highest mean MVPA observed at baseline and during the intervention	• Facilitator 0.8 (1.2) sedentary, 3.4 (2.2) moderate, 6.4 (3.7) MVPA, 3.0 (2.3) vigorous. • No facilitator 0.7 (1.0) sedentary, 3.2 (2.7) moderate, 6.2 (4.7) MVPA, and 3.0 (3.3) vigorous	-

Table 3. (Continued on next page)

Author(s), Publication Year	Sample and Setting	Setting, Research Design, Condition, and/or Stratification	SOPLAY PA Measures			Contextual Characteristics and Target Areas	Other PA Measures
			Rates of PA		MVPA		
			S, W, V				
Gustat et al. ²⁶	Children, adolescents, adults	School playgrounds (recess) - Quasi-experimental design: Intervention (walking path and playground) vs. 2 comparison neighborhoods - PACE project - Intervention; CBPR Observations (2006-2008) during weekend and weekdays (Thu.-Sun) for 6 weeks (Oct.-Dec)	Modified SOPLAY activity codes - Unclear how instrument was modified, however activity codes seem to align (S, W, V) - Increases in in both moderate and vigorous activity combined from 36.7% to 41.0% and in observed vigorous activity from 10.5% to 13.7%	b41% were moderately to vigorously active in the section of the intervention neighborhood with the walking path compared with 24% and 38% in the comparison neighborhoods at the post-intervention measurement (<i>p</i> < .001)	-	Self-report PA survey	
Huberty et al. ⁶⁰	Children 4.3-13.1 years	12 ASPs - Participants ranged from 23 to 155 children; 9:1 children-to-staff ratio - Quasi-experimental design (baseline: 1 week (Mon-Thu.); Follow-up: 3 months (Mon-Thu.) Observations (Sept.-Nov. 2010)	Standard SOPLAY activity codes (S, W, V) - Males (S/W/V) 50.5/31.4/18.2% - Females 62.1/24.0/37.9%	W + V = MVPA - Males 49.5% Females: 37.9% - Positive relationship between number of positive features and % of boys/girls engaged in MVPA - Increases in MVPA were observed when staff were directly engaged in PA, verbally promoted PA, and when PA was organized, and equipped. When 3 or more of these characteristics were present, the % children engaged in MVPA increased by 25-30%	When all features were observed the proportion of girls in MVPA doubled (30.1% vs 62.2%) and the proportion of boys engaged in MVPA increased by 25% (41.7% vs 67.1%)	-	
Huberty et al. ⁶¹	Children 7-12 years 3rd-6th grade (N = 667)	School playgrounds (recess) 12 elementary schools - Ready for Recess Intervention - Quasi-experimental design (Baseline: Aug./Sept. 2009) vs. (Post-April/May 2009-2019) 60 observations, 5 weekdays	Standard SOPLAY activity codes (S, W, V) No significant main effects for treatment condition, time, or treatment condition by-time effects for boys' and girls' sedentary and vigorous activity	V = MVPA	-	Accelerometer: ActiGraph GT1M	

Table 3. (Continued on next page)

Author(s), Publication Year	Sample and Setting	Setting, Research Design, Condition, and/or Stratification	SOPLAY PA Measures		Contextual Characteristics and Target Areas	Other PA Measures
			Rates of PA			
			S, W, V	MVPA		
Hughey et al. ⁶²	Children 5-12 years (N = ~500)	4 ASPs 10:1 children-to-staff ratio - Quasi-experimental design: pre-assessment and post-assessment - Compared ASPs (2011) and ASPs (2012)	Standard SOPLAY activity codes (S, W, V) Boys and girls also reduced sedentary behaviors at post-assessment throughout the program (–11.4%) and (–9.7%) and during scheduled PA time (–9.4% and –9.0%), respectively	V= MVPA - From baseline, 16.6% and 11.7% of boys and girls, respectively, were engaged in MVPA during scheduled PA time. At post-assessment, boys and girls engaged in MVPA during scheduled PA time increased to 20.0% and 17.3%, respectively. - Throughout the entire program, boys and girls increased MVPA levels by 4.9% and 5.1%, respectively.	The largest changes in both MVPA and sedentary behavior for boys and girls occurred during free-play PA, organized PA, and enrichment.	-
Khorana et al. ⁶³	Children 2nd-3rd grade 82-100% FRL	School playgrounds (recess) 14 elementary schools - Quasi-experimental Matched control design - WITS intervention (7 schools; 7 control schools)	Standard SOPLAY activity codes (S, W, V) - Modified 14 types of activities - Wellness in the schools (WITS) increased PA in indoor recess, a higher percentage of WITS girls and boys were vigorously active, a lower percentage were sedentary, and overall recess PA levels were higher ($p < .001$)	-	For PA type, WITS intervention schools had higher overall PA level during ball-like activities during outdoor recess in girls ($p = .002$) and boys ($p = .010$)	-
Long et al. ⁶⁴	Children 7th-8th grade (N = 5882) 17% FRL	School playgrounds (recess) Quasi-experimental modified reversal design (Alternating baseline intervention phases) observations (2014-2016)	Standard SOPLAY activity codes (S, W, V) Proportion of student observed in MVPA and sedentary behavior (%) by time	^a The proportion of students observed in MVPA increased 11% ($p < .001$), with a corresponding reduction in sedentary behavior of 11%	-	Pedometer: Ommion Altiva Self-report health behavior survey

Table 3. (Continued on next page)

Author(s), Publication Year	Sample and Setting	Setting, Research Design, Condition, and/or Stratification	SOPLAY PA Measures			Contextual Characteristics and Target Areas	Other PA Measures
			Rates of PA		MVPA		
			S, W, V				
Lorenz et al. ¹⁸	Adolescents 7th-8th grade (N = 1452) 17% FRL	School leisure-time PA - Quasi-experimental, 4 phase hybrid reversal design with alternating baseline and intervention phases - Middle school observations (Wed. and Fri.) with 3 observational lunch periods	Standard SOPLAY activity codes (S, W, V) Mean (SD) of total number of observed girls and boys in MVPA across phases and areas (least squared means)	W + V = MVPA - General linear models that regressed the # of girls and boys in MVPA on phase, activity area, and contextual variables produced model estimates that accounted for 55% of the variance in MVPA participation by girls ($p < .0001$) - No differences in the number of observed students in MVPA between the environmental and environmental+ behavioral for girls ($t_1 = -.99, p = .32$) or boys ($t_1 = -1.14, p = .26$).	Boys participated in MVPA than girls across phases, and the intervention phases were higher than baseline for girls ($F(2,1173) = 13.5, p < .0001, \eta^2 = .02$) and boys ($F(2,1173) = 20.1, p < .0001, \eta^2 = .03$).	-	
Loukaitou-Sideris and Sideris ⁶⁵	Children, adolescents 10-13 years (N = 1245)	100 Parks - Examined the utilization of the parks by SES, park attributes and demographics - Observations made (May-Jun, 2007) to compare 2 groups (inner-city, valley)	Modified SOPLAY activity codes	-	-	Did not analyze PA data Self-report PA surveys from 12 middle schools	
Mayfield et al. ⁶⁶	Children 3rd-5th grade 41% FRL	School playgrounds (recess) 4 elementary schools - Longitudinal, cluster-randomized design (2 intervention, 2 control schools) 2 waves of observations (Apr-May in 2013-2015)	Standard SOPLAY activity codes (S, W, V) % of children engaged in sedentary and MVPA by time and condition	V = MVPA - Main effects indicated no significant differences in MVPA between intervention and control schools. - Boys demonstrated greater absolute increases in MVPA	-	EER (kcal/kg/m²) Semi-structured interviews SOCARP	
McKenzie et al. ⁷	Adolescents 6-8th grade 39% FRL	School leisure-time PA 24 middle schools 3 obs. days per school 20 weeks (72 days) - Observations occurred across 3 time periods (before-, lunchtime, after-school) % MVPA by gender and time-period	Standard SOPLAY activity codes (S, W, V) Both time-period (before, lunch, after-school) and gender differences in area usage were significant	W + V = MVPA Across all measurement periods, boys in areas engaged in 64.1% vs. 48.3% MVPA and had engaged in S 36.8% of the time and engaged in W and VPA 36% and 27.2% of the time, respectively. Girls were sedentary more often than boys and they engaged in vigorous PA and MVPA less often.	- The most prominent activity that both girls and boys engaged in was coded as "no identifiable sport, game, or exercise." - For girls, this was coded 94% of the time before school, 73% at lunch time, and 87% after school. - For boys, these activities were coded 70% before-school, 37% at lunch time, and 84% after school.	EER (kcal/kg/m²)	

Table 3. (Continued on next page)

Author(s), Publication Year	Sample and Setting	Setting, Research Design, Condition, and/or Stratification	SOPLAY PA Measures		Contextual Characteristics and Target Areas	Other PA Measures
			Rates of PA			
			S, W, V	MVPA		
McKenzie et al. ¹⁶	Children (N = 36,955)	School leisure-time PA 13 elementary schools 18 months, 65 observation days from 2002 to 2004 - Observations occurred across 3 time periods (before-, lunchtime, after-school)	Standard SOPLAY activity codes (S, W, V) - Males (S/W/V): 33.7/36.5/24.6% - Females: 40/35.4/29.8% - All: 36.8/36/27.2%	W + V = MVPA Children tended to engage in more MVPA during lunch periods than during recess or before school	- Children engaged in less MVPA when areas were supervised or had organized activities. - Males were more active than females, regardless of contextual characteristics - A greater % of children were observed W and MVPA in areas not supervised	-
Moore et al. ¹⁷	Children 2nd-5th grade 53%	School leisure-time PA - Quasi-experimental design: 5 elementary schools; 3 Comprehensive School Physical Activity Program (CSPAP) and 2 control schools - Observations (2015-2016) 1 observation day per school	Standard SOPLAY activity codes (S, W, V) Examined % S and MVPA by group (CSPAPs vs controls) and setting (class, recess, PE, ASP)	^a Statistically nonsignificant difference in the % of boys and girls engaged in MVPA was observed between CSPAP schools and control schools, with higher levels in the control schools**	No differences were seen in the percentage of children engaged in MVPA or sedentary behaviors in PE class between groups, nor did the instructional practices differ.	Accelerometer: ActiGraph GTX3+ SOSPAN SOSMART
Raney et al. ⁶⁷	Children 1-5th grade (n = 437)	School playgrounds (recess) 2 elementary schools; 4 playgrounds - Nonrandomized control trial (experimental vs control at baseline and follow-up) - Observations (2016-2017) 2 weeks prior (pre-greening), immediately after (greening), 4 months after (follow-up)	Standard SOPLAY activity codes (S, W, V) - Examined % (S, W, V) by condition, location (pre-greening, post-greening, follow-up), and surface type (hardscape, green space) - The students' PA level was significantly affected by location and study phase.	-	- Children highest % sedentary occurred in handball, kickball, and volleyball areas independent of study phase, location or sex - Regardless of sex, more experimental students were active in green space compared with hardscape areas (F(1, 2282) = 7.13, p < .001)	Accelerometer: ActiGraph GT3X+ SOCAPP
Robinson et al. ³⁴	Children 3-5 years Pre-K (N = 97)	School playgrounds (recess) 2 Head Start pre-schools - Quasi-experimental repeated-measures design: - Experimental intervention (Arm I) and a combination of environmental & educational intervention (Arm II) 6 weekday observations; 1 observation day per school	Standard SOPLAY activity codes (S, W, V) The % of students observed as sedentary decreased by 10.0% at 4-months, vigorous increased pre- to post-assessment at the individual and population level and remained higher than pre-greening at 4-months for girls in the experimental group only	-	- No change in light, moderate, or vigorous PA in preschool children following the provision of equipment - Zone popularity and recess PA behaviors did not change for control students during the study (p > .05)	Accelerometer: ActiGraph GT3S

Table 3. (Continued on next page)

Author(s), Publication Year	Sample and Setting	Setting, Research Design, Condition, and/or Stratification	SOPLAY PA Measures				Other PA Measures
			Rates of PA		Contextual Characteristics and Target Areas		
			S, W, V	MVPA			
Saint-Maurice et al. ⁶⁸	Children 8-12 years 3rd-5th grade (N= 100)	School playgrounds (recess) 2 elementary schools - Quasi-experimental design: experimental ready for recess intervention; baseline, trial 1, post-assessment, and trial 2 3 consecutive weekdays (Tue-Thu) - Observations (Fall 2008-Spring 2009)	Standard SOPLAY activity codes (S, W, V) - SOPLAY = PA levels x Features - Examined MVPA (%) by gender, race, grade, BMI, and group - Sedentary behavior by gender and target area interaction was nonsignificant; $F(6,219) = 1.45$, $p = .20$	V = MVPA Males: 34.1% Females: 22.4%	- Males: 41.9% - Females: 43.9% (27.0) - The MVPA by gender and area interaction was not significant ($F(6,219) = 1.21$, $p = .303$), but boys were more active than girls in 5 of 7 target areas - Zone A had the highest % of boys and girls engaged in MVPA	Accelerometer: ActiGraph GT1M	
Saint-Maurice et al. ⁶⁹	Children 9-12 years 4th-6th grade (N= 160)	School playgrounds (recess) 9 elementary schools & 9 ASP - Validation study - Observations (Fall 2008-Spring 2009)	Standard SOPLAY activity codes (S, W, V) Examined mean differences between accelerometers and SOPLAY classifications (S, W, V) for validation purposes	W + V = MVPA - Males: MVPA 1-73.2% MVPA 2-23.8% MVPA 3-73.2% - Females: MVPA 1-73.15% MVPA 2-23.32% MVPA 3-73.15% - All: MVPA 1-72.64% MVPA 2-23.42% MVPA 3-72.64%	-	Accelerometers: Biotrainer Pro Activity	
Sallis et al. ¹⁹	Adolescents 6th-8th grade 39% FRL	School leisure-time PA 24 public middle schools 3 observation days per school	Standard SOPLAY activity codes (S, W, V) - Examined % of S, W, V by various environmental variables - The proportion of girls observed being physically active ranged from 0% to 5% across schools, with a mean of 1.6% - The proportion of physically active boys ranged from 1% to 11%, with a mean of 5.5%	-	- The interaction area type by equipment indicated that more girls were active when equipment was not present in indoor areas, but equipment enhanced activity levels in outdoor areas - The interaction supervision by area improvements indicated that girls were most active when the environments had high levels of improvements and supervision	MVPA score converted to EER - (kcal/kg/m ²)	

Table 3. (Continued on next page)

Author(s), Publication Year	Sample and Setting	Setting, Research Design, Condition, and/or Stratification	SOPLAY PA Measures			Other PA Measures
			Rates of PA		Contextual Characteristics and Target Areas	
			S, W, V	MVPA		
Sallis et al. ²⁰	Adolescents 6th-8th grade 39.5% FRL	School leisure-time PA • 24 middle schools • RCT; middle school physical activity and nutrition (M-SPAN) intervention • Compared 12 intervention (M-SPAN) and 12 control schools; Spring 1997-1999 • Baseline: 3 random days per school; Intervention: 2 random days at each school per semester during 4 time periods (PE classes, before-, lunchtime, and after-school)	Standard SOPLAY activity codes (S, W, V) • Examined % of children in MVPA by area type (court, field, indoor) and levels of supervision • Intervention schools increased PA over time at a greater rate • Gender-specific secondary analyses revealed the time by condition interaction was significant for boys, with a large effect size. The interaction was not significant for girls, and the effect size was small.	-	-	SOFIT Student and parent self-report surveys MVPA score to EER • (kcal/kg/m ²)
Smith et al. ¹⁵	Children 6-12 years K-6th grade	School leisure-time PA • 4 elementary schools • 30 days of observations; each school observed weekdays (Monday-Friday) during 3 periods (before school, lunchtime, after-school) across 6 weeks	Standard SOPLAY activity codes (S, W, V) • Examined PA % S, W, V stratified by gender, intensity and time • Before School (S/M/V): Males: 39.7/45.6/50.2% Females: 44.5/48.3/37.2% • Lunchtime (S/M/V): Males: 69.8/70.3/84.4% Females: 76.7/73.6/26.2% • After School (S/M/V): Males: 35.4/26.2/9.1% Females: 37.20/26.2/6.10%	W + V = MVPA • Both sexes displayed a fewer number of walking and very active PA counts compared to sedentary counts during the after-school leisure time	-	-
Spengler et al. ⁷⁰	Children < 10 years, adolescents, adults (N = 3410) Tampa (n = 2358) Chicago (n = 1052)	Parks • Cross-sectional design • 28 public parks; 18 Chicago and 10 Tampa Parks • 4 observation periods: 2 AM, 2 PM	Standard SOPLAY activity codes • Examined % S, W, V by characteristics, gender and race/ethnicity • Children (S/M/V) Tampa park: 55.6/26.6/17.9% Chicago park: 48.0/27.7/24.3% • Adults (S/M/V) Males: 55.7/26.2/18.1% Females: 55.4/27.1/17.5%	^a Overall, 44.5% of children in Tampa engaged in MVPA, whereas 55.5% of Chicago participated in MVPA when they were observed**	• 5 target areas Court: 23.9/50.7/25.4% Field: 59.5/25.0/15.5% Playground: 45.6/28.4/26.0% Shelter: 75.3/198/4.9% Open space: 47.2/31.4/21.5%	SOPARC

Table 3. (Continued on next page)

Author(s), Publication Year	Sample and Setting	Setting, Research Design, Condition, and/or Stratification	SOPLAY PA Measures		Contextual Characteristics and Target Areas	Other PA Measures
			S, W, V	MVPA		
Stanton-Chapman et al. ⁷¹	Children 5-12 years (N = 2296)	School playgrounds (recess) - Examined differences between 2 playgrounds - Weekdays and weekend 30 days at Playground A; 29 days at Playground B	Standard SOPLAY activity codes (S, W, V) - Children tended to engage in W behaviors more than V - Statistically significant differences by area for all children, but no statistically significant differences by area for sex for 2 playgrounds	-	- Females: areas differ in terms of S [F (4, 60) = 3.88, $p < .01$], W [F (4, 60) = 8.78, $p < .001$], V [F (4, 60) = 5.13, $p < .01$] - Males: significant differences for S [F (4, 60) = 4.53, $p < .01$], W [F (4, 60) = 8.01, $p < .001$], V [F (4, 60) = 6.32, $p < .001$]	METs
Suau et al. ⁷²	Children, adolescents, and adults (N = 9452) Tampa (n = 7112); Chicago (n = 2402)	29 Parks - 10 Tampa; 19 Chicago - Observational (Direct) design 4 observation periods - Weekdays (Fri-Sun.) - Observations during Spring (Tampa) and Summer (Chicago) 2005	- Modified SOPLAY activity codes - Overall, METs generated by Tampa parks corresponded to sedentary and light-intensity activities, whereas METs for Chicago parks corresponded on average to moderate-intensity activity levels	-		METs
Weaver et al. ⁷³	Children <12 years	4 SDCs - Experimental design (pre- and multiple post-) 67 program days in the Summer 2011 (ASP 1) and Summer 2012 (ASP 2) - Stratified PA by target areas, staff behaviors and child PA levels	Standard SOPLAY activity codes (S, W, V) - Examined % S and MVPA of children observed - Males: Change ranged from 11.6% decrease to a 6.9% decrease observed in S; Changes in % of boys in MVPA and sedentary was consistent across grade levels - Females: Change ranged from 12.0% decrease to 4.2% decrease observed in S; Changes in % of girls in MVPA and sedentary fluctuated across grade levels	V = MVPA - Males: Change ranged from 6.2% increase to a 3.5% increase - Females: Change ranged from 7.6% increase to a -0.1% decrease % of girls and boys observed in moderate-to-vigorous physical activity increase from 15.3% to 18.3% ($p > .05$) and 17.9% to 21.2%, whereas sedentary behavior decreased from 66.8% to 59.8% and 62.3% to 53.6%, respectively	% of boys and girls engaged in sedentary and MVPA by scheduled activity measures based on SOSPAN and SOPLAY scans	SOSPAN

Table 3. (Continued on next page)

Author(s), Publication Year	Sample and Setting	Setting, Research Design, Condition, and/or Stratification	SOPLAY PA Measures			Contextual Characteristics and Target Areas	Other PA Measures
			S, W, V	Rates of PA	MVPA		
Yu et al. ⁷⁴	Adolescents 6th-8th grade (N = 514)	School playgrounds (recess) • 2 middle schools • Baseline-intervention design • Baseline phase followed by equipment provision intervention phase • Mean % of MVPA x School x Gender x Group	Standard SOPLAY activity codes (S, W, V) • Examined the mean % of MVPA by school (1 and 2), gender (male and female), and phase	W + V = MVPA • Overall % of MVPA levels increased in both schools • Average MVPA % in each school increased during the intervention in all grades. • In school 1, the total % of MVPA increased 14%. Total % of MVPA levels increased in school 2, along with significant differences reported with 7th-8th grade students MVPA %	-	-	
Zarrett et al. ³²	Children 5-17 years	Summer day program • 4 CDCs; 2 high-resourced • Observational (direct) design • 4 observation days (2 weeks) • Examined %S, W, V and stratified by intensity, physical features, staff interactions, and social climate components	Standard SOPLAY activity codes (S, W, V) • S/W/V: 72/13.8/13.8%	^a Across camps, 28% of instances youth were involved in MVPA • Males: 30% • Females: 23%	• Presence of equipment and higher temperatures was related to higher MVPA participation rates for females • Physical features of the camp setting were not predictive of PA when examined across the full sample	EER • (kcal/kg/m²) converted to METs	
Zarrett et al. ³¹	Children 7-12 years K-12th grade 90% FRL	ASPs • PA stratified by intensity, physical features, staff interactions, and social climate components • Observations occurred over 5 weekdays (Monday through Friday) across 2 weeks	Standard SOPLAY activity codes (S, W, V) • 72.6% sedentary, 17.8% walking, 9.6% very active • Boys were more active than girls with 33.5% (12.4% vigorous) of boys observed freestyle involving MVPA as compared with only 18.9% (5.7% vigorous) of girls' scans	^a Across ASPs, youth were involved in MVPA in 27.4% (9.6% vigorous) of observed scans**	• All physical-environmental features and controlling for variations by ASP, indicated significant independent contributions of organized activities, $\beta = 3.45$, $t(583) = 2.01$, $p = .05$, and equipment, $\beta = 3.41$, $t(583) = 2.38$, $p = .02$—on youth PA across the sample	-	

PA, physical activity; MVPA, moderate-to-vigorous physical activity; S, Sedentary (low), W, Walking (moderate), V, Very Active (vigorous), LL, learning landscape; EER, estimated energy expenditure rates; METs, metabolic equivalents; PARA, physical activity resources assessment, FRL, free or reduced lunch, ASP, afterschool program, RCT, randomized controlled trial.

^a Somewhat unclear how SOPLAY activity codes were combined to report MVPA;

^b Clear how SOPLAY activity codes were used, however, MVPA not reported.

percentage of MVPA compared to girls during school recess.⁷⁷⁻⁷⁹

An advantage of SOPLAY is that researchers can modify specific elements (eg, activity type and contextual characteristics) while evaluating group-level PA as compared to wearable devices. While the purposes of the studies differed substantially, we suggest future investigations reduce the variability in how PA is reported (eg, METs, individual level data, use of different activity codes, etc.). While assessing MVPA, only a few authors provided citation of a SOPLAY validation study,⁶⁹ wherein MVPA is calculated with the “W” and “V” codes. Otherwise, authors should be clear in their use of activity codes. Using the methods prescribed in the original SOPLAY protocol to assess PA and study contexts would be beneficial in making comparisons across studies.

Contextual characteristics. Most reviewed studies described collecting data at school playgrounds, neighborhood parks, after-school programs, summer day camps, and other settings (eg, museum). Notably, some settings were not clearly described. Several studies appeared to be conducted during recess, however, this information was not explicitly stated and subsequently not coded. Importantly, approximately 90% of the studies did not include the standardized contextual information used in the SOPLAY instrument (ie, accessible, usable, supervised, organized, equipped) when considering PA participation. This contextual information related to play and leisure settings is a strength of SOPLAY so contextual variables should be reported for both academic and practical purposes (eg, does adding equipment or direct supervision affect PA).

Inconsistency with implementation practices. A major finding of our review is that studies often deviated from the original SOPLAY protocol. For example, they sometimes defined terms/concepts in different ways, provided insufficient information about study methods including observer training and interobserver reliability, and failed to clearly report about the context and PA. The omission of such information makes it difficult to determine the consistency of procedures and the quality of studies. As mentioned, it also challenges comparisons across studies being made. Adaptations to the SOPLAY instrument may be appropriate for local studies, but in order for generalizations to be made to other studies, researchers need to follow the original protocol for observer training and data collection (available online; <https://activelivingresearch.org>).⁶ Moreover, authors should report all descriptive information for the studies (eg, means, standard deviations, frequencies) for comparability across them.

Strength and Limitations

This review covers 2 decades of research of all published studies that used SOPLAY in the United

States and Canada. Articles were reviewed by multiple trained coders and attempts to summarize the key findings from the studies that met the established eligibility criteria. Regardless of reported results, the articles exemplify the broad range of research aims, designs, and methods for analyzing variables used with SOPLAY.

There were notable limitations of this review. First, the studies included were limited to original research conducted in North America and published in English. The inclusion of only North American countries limits the utility and applicability of the findings more broadly. Second, the authors did not perform a risk of bias assessment of included studies, thus study results were not ranked, weighted, or grouped according to quality. Due to the heterogeneity of study characteristics (ie, participants observed, intervention [settings] selection, outcomes assessed, and study design), statistical analysis was limited. The methodologies restricted reporting of statistics, which made comparisons and summaries of articles problematic. Confronted with this issue, the authors were limited to statistical analysis performed with key variables, for instance only 35 studies reported MVPA. As a result, there were a lack of studies measuring and analyzing outcomes variables in any given category (eg, PA, contextual, target area) with similar participants, interventions, and designs to provide a summary statement and report given magnitudes of effects sizes.

Third, this review focused on examining an instrument with the capability of evaluating group-level PA. Since the development of other direct observation tools recently, researchers interested in youth PA now have the possibility to capture behavioral data at the individual-level. Another limitation of this study was failure to conduct quality assessments for each study, as the authors found criterion outlined by the Effective Public Health Practice Project and Cochrane’s Risk of Bias-2 tool insufficient for evaluating nonclinical trial studies included in this review. Lastly, following the completion of this systematic review, at least 1 study examined youth PA behavior using SOPLAY in the United States was published.⁷⁴ Another study adapted SOPLAY to observe interactions between child PA, shade and nature during recess.⁸⁰ These few studies add to the evidence suggesting results concerning children and adolescent PA behaviors and contextual characteristics associated with leisure-type contexts were mixed.

Recommendations for Future Studies Using SOPLAY

Following a standard protocol for SOPLAY observer training and data collection is critical to ensuring procedural consistency and the generalizability of

findings. The use of specific terms should also be consistent. For example, the word “observation” stands out. In the reviewed studies, the term “observation” was used in a diverse and inconsistent manner (eg, from describing an entire day of data collection to data on a single individual). We recommend using more precise terms, such as (a) “observation days,” (2) “observation sessions” that would more precisely identify the number of sessions made before, during, and after school, (3) “observation scans” to identify the number of times a target area was scanned using the prescribed moving from left to right protocol in which each person is recorded, and (4) and “individuals observed” to identify the number of people in the setting.

Future researchers should precisely explain the procedures used (eg, exact setting, observer training methods) and report observer interrater reliability data. In addition, all the data collected on the SOPLAY instrument should be reported (eg, MVPA, contextual data) when relevant. It should be noted that not all contextual data is relevant to all studies. As only SOPLAY studies in North America were reviewed, additional reviews of studies in other locations are recommended.

IMPLICATIONS FOR SCHOOL HEALTH

The results of this review support strategies and initiatives that schools can implement to assist students in meeting national PA recommendations. First, substantial evidence suggest that youth accrue a significant proportion of PA during leisure-time opportunities, therefore, health and educational professional interested in health-related outcomes (ie, MV[PA]) should consider utilizing SOPLAY. Objective evaluations of play and leisure time areas both within and beyond school settings are needed to better understand environments and to help increase PA engagement that leads to improved health and wellness. Apart from evidence of students engaging in PA during physical education class,^{81,82} additional research in the area leisure and play settings is warranted.

Consequently, SOPLAY instrument is a valid and reliable systematic observation tool designed to reach this end. Results suggest that SOPLAY is a versatile and precise instrument that allows one or more individuals to analyze youth PA within groups in varied unstructured PA contexts. It also provides important contextual information to better understand leisure PA and play opportunities. Furthermore, the evidence from this review suggests that SOPLAY is an effective instrument to measure PA in leisure-time and play settings, such as recess and before- and after-school programs, parks and playgrounds for all school-aged children. This instrument can be used to assess group level PA information without human subject's approval, providing practical value for research as well

as other types of activities (eg, field trips, wellness programs). Measuring student PA behavior during school, such as recess can become complicated and costly, whereas individuals may become trained to use SOPLAY in a short time and at a limited cost.

In sum, the results from this review point toward the importance of contextual characteristics, such as available playground equipment, access to locations, organized PA opportunities, or supervision from staff or teachers.^{34,59,60} SOPLAY may assist those responsible with understanding the school and community areas that are available for children and adolescents to engage in leisure PA throughout the day. As an effective tool for measuring PA and examining critical features of the school environment, studies included in this review highlight the importance of understanding factors influencing child and adolescent PA. Findings from these studies suggest the need to explore the complex interactions between the social and environmental determinants of PA during leisure PA and play opportunities further. Development of school physical activity policies should be supported with evidence from objective, feasible methods. Schools should consider utilizing SOPLAY as a procedure for gathering and disseminating data to promote beneficial physical activity and health and wellness policies.

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SUPPORTING INFORMATION

The following Supporting Information is available for this article:

Data S1: Keyword Search Algorithm

Additional supporting information may be found online in the Supporting Information section at the end of the article.